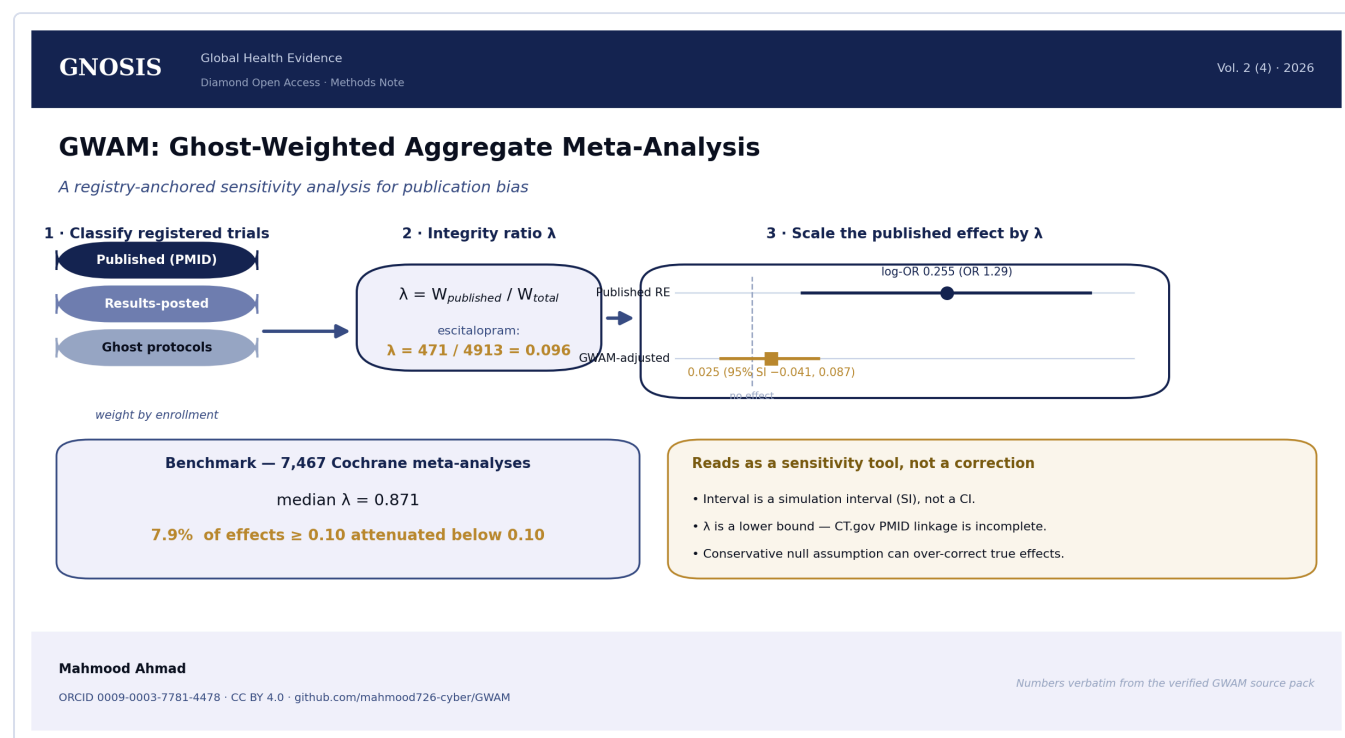


GWAM: Ghost-Weighted Aggregate Meta-Analysis for registry-based publication-bias sensitivity

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Visual abstract. Registry classification → integrity ratio λ → λ -scaled effect, with the escitalopram application and the Pairwise70 benchmark. All values are verbatim from the verified GWAM source pack.

ABSTRACT

Background. Publication bias threatens the validity of meta-analysis, yet routine detection tools (funnel-plot asymmetry, trim-and-fill) rest on assumptions that are hard to verify. Trial registries offer an external anchor for how much of the evidence base is actually visible.

Methods. GWAM (Ghost-Weighted Aggregate Meta-Analysis) is a registry-anchored sensitivity method. It classifies ClinicalTrials.gov records as ghost (completed, with no publication and no posted results), results-posted, or published (PMID-linked); computes an enrollment-weighted integrity ratio λ (the published-participant fraction); and scales the published pooled effect by λ . A Monte-Carlo layer and a hierarchical variance-propagation extension carry the uncertainty about unobserved trials.

Results. For escitalopram in depression (18 trials) λ was 0.096; the GWAM-adjusted log-odds-ratio was 0.025 (95% SI -0.041 to 0.087; SI denotes a simulation interval, not a confidence interval) versus a published random-effects log-odds-ratio of 0.255 (OR 1.29). Across a benchmark of 7,467 meta-analyses from 398 Cochrane reviews the median λ was 0.871, and 7.9% of meta-analyses with a random-effects estimate of magnitude at least 0.10 were attenuated below 0.10. For pregabalin (35 trials, log-risk-ratio) λ was 0.014, flagged as likely incomplete PMID linkage. In simulation (2,000 replications) bias was minimal at $\mu=0$ but grew negative for $\mu>0$ (-0.162 at $\mu=0.2$) and raw coverage fell to 7.1%; the hierarchical extension restored near-nominal coverage.

Conclusions. GWAM is a transparent, reproducible registry-anchored sensitivity analysis, best interpreted as a conservative framework that complements existing publication-bias methods; its performance depends on the quality of ClinicalTrials.gov publication linkage.

PIPELINE

Fig 1. GWAM Pipeline Overview

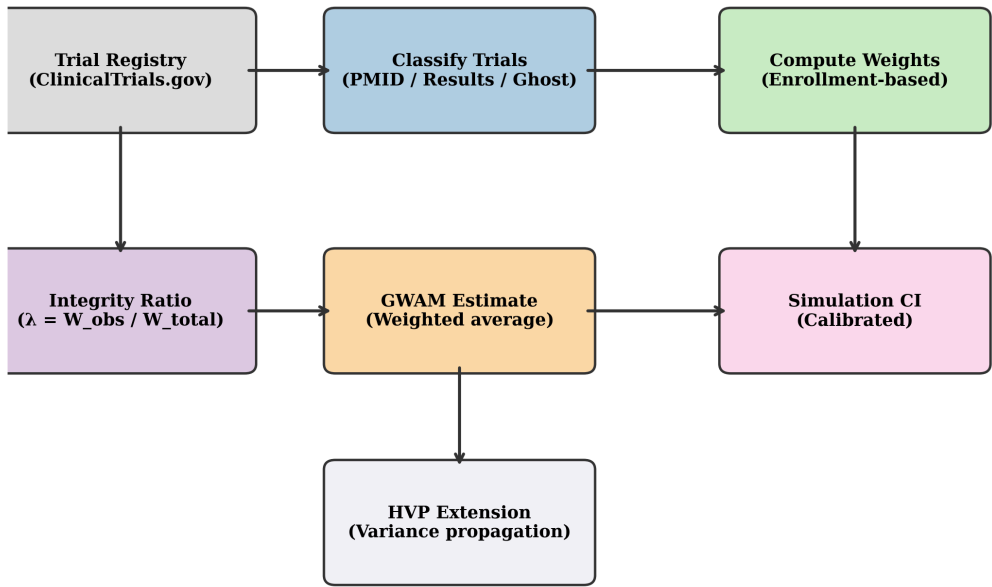


Figure 1. GWAM pipeline overview. The seven-stage workflow runs from the ClinicalTrials.gov registry query through trial classification (published / results-posted / ghost), enrollment-weight computation, integrity-ratio (λ) calculation, GWAM point estimation, Monte-Carlo simulation-interval construction, and the optional hierarchical variance-propagation extension.

TABLE 1 — INTEGRITY RATIO (λ) AND GWAM ATTENUATION

Panel	Stratum / drug-condition	n	λ	Published effect (log scale)	GWAM result
Applications	Escitalopram / depression	18 trials	0.096	0.255 (OR 1.29)	0.025 (95% SI -0.041, 0.087)
Applications	Pregabalin / neuropathic pain	35 trials	0.014	0.916 (RR 2.50)	0.013 (95% SI -0.029, 0.058)
Benchmark (overall)	All Cochrane meta-analyses	7,467	0.871 (median)	median $ \Delta $ 0.055	7.9% attenuated <0.10 (95% CI 5.1–11.0)
Benchmark by λ source	Exact NCT linkage	1,438	0.950	—	3.1% attenuated <0.10
Benchmark by λ source	CT.gov query linkage	5,569	0.861	—	9.1% attenuated <0.10
Benchmark by λ source	Transport proxy	86	0.950	—	10.5% attenuated <0.10
Benchmark by λ source	Default (0.70)	374	0.700	—	8.3% attenuated <0.10
Benchmark by study count	k = 2	2,895	0.894	—	6.3% attenuated <0.10
Benchmark by study count	k = 3	1,409	0.880	—	7.9% attenuated <0.10
Benchmark by study count	k ≥ 4	3,163	0.845	—	9.4% attenuated <0.10

λ = enrollment-weighted integrity ratio (published-participant fraction); for applications λ is the directly measured λ_{pmid} , for the benchmark it is the proxy λ . Effects are on the log-OR (escitalopram) or log-RR (pregabalin) scale; positive values indicate treatment superiority. SI = simulation interval (not a confidence interval). “% attenuated <0.10” = share of meta-analyses with $|\text{random-effects estimate}| \geq 0.10$ whose GWAM-adjusted $|\text{estimate}|$ fell below 0.10. Δ = GWAM – random-effects shift. Pregabalin’s $\lambda = 0.014$ reflects incomplete ClinicalTrials.gov PMID linkage, not true non-publication.

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Honest-reporting note. The 95% interval on the GWAM-adjusted estimate is a **simulation interval (SI)**, not a confidence interval. The integrity ratio λ is a **lower bound** on the true publication fraction because ClinicalTrials.gov PMID linkage is incomplete; GWAM is a conservative **sensitivity** framework that complements, and does not replace, existing publication-bias methods.